# Yasuhiro Sato, Ph.D.

Last update on 19 September 2026

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## Curriculum Vitae

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Google scholar citations: https://scholar.google.co.jp/citations?user=HbrGYr8AAAAJ&hl=en
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### Education

- 23 March 2016. PhD in Science (Biosciences), Center for Ecological Research, Graduate School of Science, Kyoto University, Japan (supervisor: Prof. Dr. Hiroshi Kudoh) [Thesis title: Associational anti-herbivore defense in the trichome dimorphism of *Arabidopsis halleri* subsp. *gemmifera* (Brassicaceae); available at https://doi.org/10.14989/doctor.k19536]
- 25 March 2013. MSc in Science (Biosciences), Center for Ecological Research, Graduate School of Science, Kyoto University, Japan (supervisor: Prof. Dr. Hiroshi Kudoh)
- 24 March 2011. BSc in Agriculture (Bioresource Sciences), Laboratory of Insect Ecology, Faculty of Agriculture, Kyoto University, Japan (supervisor: Assistant Prof. Dr. Takayoshi Nishida)

### Employment history

- April 2024–Present. Tenure-track Assistant Professor in the Group of Ecological Genetics, Section of Environmental Biology, Faculty of Environmental Earth Science, Hokkaido University, Japan
- April 2021–March 2024. Senior research and teaching assistant (Oberassistent) at the Department of Evolutionary Biology and Environmental Studies, University of Zurich, Switzerland (PI: Prof. Dr. Kentaro K. Shimizu)
- October 2017–March 2021. Junior Group Leader of Precursory Research for the Embryonic Science and Technology (PRESTO) program of the Japan Science and Technology Agency (JST) at Ryukoku University, Japan (host: Associate Prof. Dr. Atsushi J. Nagano)
- April 2016–September 2017. Independent Postdoctoral Fellow of the Japan Society for the Promotion of Science (JSPS) at Ryukoku University, Japan (host: Senior Lecturer Dr. Atsushi J. Nagano)
- April 2015–March 2016. Research Fellow of the Japan Society for the Promotion of Science (JSPS) for PhD students at Kyoto University, Japan (host: Prof. Dr. Hiroshi Kudoh)
- June 2013–March 2014. Research Assistant at the Center for Ecological Research, Kyoto University, Japan

## Visiting and guest position

- October 2017–March 2024. Guest researcher of Research Institute for Food and Agriculture, Ryukoku University, Japan
- August 2017–December 2024. Affiliated member of University Research Priority Program for Global Change and Biodiversity (URPP GCB), University of Zurich, Switzerland https://www.gcb.uzh.ch/en/aboutus/people.html
- June 2016–September 2016. Visiting postdoctoral researcher, Evolutionary and Ecological Genomics Group (Shimizu Lab), University of Zurich, Switzerland
- April 2016–March 2020. Affiliated scientist of the Center for Ecological Research, Kyoto University, Japan

## Approved funding and fellowships

1. April 2025–March 2029. 'From varieties to variety mixtures: measuring and modeling interactions and developing methods for optimizing variety combinations', Grant-in-aid for Scientific Research (A), Grant ID:JP25H00928, Japan Society for the Promotion of Science (JSPS), Japan (co-investigator; direct cost 3,200,000 JPY + indirect cost 1,200,000 JPY)
2. October 2024–March 2028 [Phase I]. 'Establishment of quantitative ecological genetics for the control of plant-plant interactions', Fusion Oriented REsearch for disruptive Science and Technology (FOREST), Japan Science and Technology Agency (JST), Japan (principal investigator; direct cost 50,000,000 JPY + indirect cost 15,000,000 JPY)
3. November 2023–October 2024. 'Using open data to reveal the genomic basis that underpins positive plant-plant interactions', Spark rapid funding, Grant ID:CRSK-3_221418, Swiss National Science Foundation (SNSF), Switzerland (principal investigator; total budget 100,000 CHF = 103,616 EUR upon the currency rate of the starting date)
4. April 2023–March 2027. 'Joint analysis of plant-plant interactions using field GWAS and RNA-Seq', Grant-in-aid for Early-career Scientists, Grant ID:JP23K14270, Japan Society for the Promotion of Science (JSPS), Japan (principal investigator; direct cost 3,600,000 JPY + indirect cost 1,080,000 JPY = 32,116 EUR upon the currency rate of the starting date)
5. April 2020–March 2023. 'Theoretical consideration of plant biodiversity effects driven by inter-genotypic interactions', Grant-in-aid for Early-career Scientists, Grant ID:JP20K15880, Japan Society for the Promotion of Science (JSPS), Japan (principal investigator; direct cost 3,200,000 JPY + indirect cost 960,000 JPY = 35,049 EUR upon the currency rate of the starting date)
6. October 2017–March 2021. 'Large-scale omics approach towards a pest control by mixed planting', PRESTO project, Grant ID:JPMJPR17Q4, Japan Science and Technology Agency (JST) (principal investigator; direct cost 40,000,000 JPY + indirect cost 12,000,000 JPY = 391,418 EUR upon the currency rate of the starting date + group leader's own salary) https://www.jst.go.jp/kisoken/presto/en/project/1112074/1112074_26.html
7. April 2016–September 2017. 'Genome-wide association study and prediction of pest communities on natural accessions of the model plant species', Research Fellowships for Postdoctoral Researcher, Grant ID:JP16J30005, Japan Society for the Promotion of Science (JSPS) (principal investigator; direct cost 2,700,000 JPY + indirect cost 810,000 JPY = 27,449 EUR upon the currency rate of the starting date + fellow's own salary)
8. April 2015–March 2016. 'A test of the mechanism maintaining plant defense polymorphism in a wild *Arabidopsis*', Research Fellowships for PhD Students, Grant ID:JP15J00400, Japan Society for the Promotion of Science (JSPS) (principal investigator; direct cost 1,200,000 JPY + indirect cost 360,000 JPY = 12,101 EUR upon the currency rate of the starting date + fellow's own salary)

## Mentoring

1. Iqra Akram (PhD student at Hokkaido University; Role: supervisor)
2. Lukas Rohr (Master student at the University of Zurich; Role: co-supervisor)
3. Dr. Chongmeng Xu (former PhD student at the University of Zurich, PhD defended on 17 April 2023; Role: co-supervisor as 'Leitung der Dissertation')

4. Elina Jansone (Undergraduate internship at the University of Zurich; Role: supervisor)
5. Kazuya Takeda (Research assistant at Ryukoku University; Role: daily supervisor as an employer)
6. Silvija Milosavljevic (Undergraduate internship at the University of Zurich; Role: co-supervisor)
7. Dimitri Anderfuhren (Undergraduate internship at the University of Zurich; Role co-supervisor)

## Teaching experience

1. April 2025–Present. Introduction to Environmental Earth Science, Hokkaido University, Japan (Role: co-instructor)
2. April 2024–Present. Environmental Biology II, Hokkaido University, Japan (Role: co-instructor)
3. April 2024–Present. Fundamental Course in Ecology, Hokkaido University, Japan (Role: co-instructor)
4. April 2024–Present. Natural Science Experiments, Hokkaido University, Japan (Role: co-instructor)
5. May 2024. Special Lecture of Agricultural and Environmental Biology, University of Tokyo, Japan (Role: one-time guest lecturer)
6. September 2021–March 2024. BIO373: Next Generation Sequencing for Evolutionary Functional Genomics, University of Zurich, Switzerland (Role: course management, co-instructor, and examiner; No. of participants = 15-20)
7. May 2022. BIO334: Practical Bioinformatics, University of Zurich, Switzerland (Role: temporal co-instructor during a sabbatical absence of PI; No. of participants = ca. 25)
8. April 2019–May 2019. External Lecturer of Basic Genetics, Faculty of Agriculture, Ryukoku University, Japan (Role: 90-min. lectures × five times for a liberal art course in Japanese; No. of participants = ca. 50)
9. July 2018–August 2018. International Biology Undergraduate Summer School (BUSS) 2018, University of Zurich, Switzerland (Role: supervision of undergraduate student's fieldwork and presentation; No. of participants = 1)
10. July 2017–August 2017. BIO378: Research Practical in Evolutionary Biology and Systematics, University of Zurich, Switzerland (Role: supervision of undergraduate student's fieldwork, data analysis, and report writing; No. of participants = 1)
11. September 2013–January 2015. External Teaching Assistant of Programming Practical, Faculty of Science and Technology, Ryukoku University, Japan (Role: teaching assistant of R language; No. of participants = 5-10)

## Panels, boards, and reviewing activities

1. April 2025–Present. Associate Editor of ***Plant Species Biology***
2. April 2024–Present. Organizing committee member of the poster presentation for high-school students, Ecological Society of Japan
3. January 2019–December 2024. Editorial board member of the Society for the Study of Species Biology, Japan
4. October 2023–November 2024. Guest editor for the special issue "Ecology and Evolution" in ***Plant and Cell Physiology***
5. April 2020–December 2021. Reviewing editor of ***Journal of Evolutionary Biology***
6. December 2022. Organizing committee member of the 52nd symposium of the Study of Species Biology, Japan, https://sites.google.com/view/sssb2020shiga
7. November 2016. Organizing committee member of 5th Japan-Taiwan Ecology Workshop, Kyoto, Japan, https://sites.google.com/site/jtecolw5th/
8. Total 54 peer-review activities verified by the Web of Science, https://www.webofscience.com/wos/author/rid/N-7939-2013.

## Active memberships in scientific societies

1. American Society of Naturalists (October 2017–Present)
2. Ecological Society of Japan (October 2011–Present)
3. European Society for Evolutionary Biology (December 2020–Present)

4. Japanese Society of Plant Physiologists (January 2018–Present)
5. Society for the Study of Species Biology (December 2010–Present)
6. Society of Evolutionary Studies, Japan (November 2024–Present)

## Organization of conferences

1. December 2019. **Sato Y**, Muranaka T. Cutting-edge technology of plant omics and phenotyping (in Japanese). The 51st Symposium of the Society for the Study of Species Biology, Miyazaki, Japan
2. March 2015. Sakata Y and **Sato Y**. Overview of evolutionary ecology of plant-herbivore interactions (in Japanese). Workshop 12: The 62nd Annual Meeting of the Ecological Society of Japan, Kagoshima, Japan

## Prizes and awards

1. 8 April 2026. The Young Scientists' Award, The Commendation for Science and Technology, Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan
2. 2 February 2018. Inoue Research Award for Early-career Scientists, Inoue Foundation for Science, Japan
3. 3 December 2016. Kataoka Award for Early-career Researchers, The Society for the Study of Species Biology, Japan
4. 2 March 2016. Ikushi Prize for Outstanding PhD Students, Japan Society for the Promotion of Science (JSPS), Japan https://www.jsps.go.jp/english/e-ikushi-prize/awards_fy2015_01.html
5. April 2011–March 2015. Refund Exemption for Excellent Students, JASSO Scholarship, Japan

## Career breaks

- June 2024–September 2024. Parental leave at Hokkaido University, Japan

## Personal skills

### License

1. Japanese diver license
2. Swiss driver license

### Computer languages

1. R (package developer level)
2. Python (basic level)
3. Bash (basic level)

### Natural languages

1. English (proficient)
2. Japanese (native speaker)

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# Achievements

*†Equal contribution; *Correspondence; #Mentee*

## Peer-reviewed articles

1. **Sato Y***, Nishida T. (2026) Recurrent visitation by lycaenid butterflies and honeybees to the same individuals of *Achyranthes bidentata* (Amaranthaceae). ***Plant Species Biology*** 41:e70064. https://doi.org/10.1111/1442-1984.70064

2. **Sato Y***. (2026) Negative frequency-dependent selection underlies overyielding through neighbour genotypic effects in *Arabidopsis thaliana*. ***Philosophical Transactions of the Royal Society B*** 381:20250179. https://doi.org/10.1098/rstb.2025.0179
3. Akram I#, Rohr L#, Shimizu KK, Shimizu-Inatsugi R, **Sato Y***. (2026) Spatial heterogeneity of disease infection attributable to neighbor genotypic identity in barley cultivars. ***in silico Plants*** 8:diaf023. https://doi.org/10.1093/insilicoplants/diaf023
4. **Sato Y***, Shimizu-Inatsugi R, Takeda K, Schmid B, Nagano AJ*, Shimizu KK*. (2024) Reducing herbivory in mixed planting by genomic prediction of neighbor effects in the field. ***Nature Communications*** 15:8467. https://doi.org/10.1038/s41467-024-52374-7
5. Stockenhuber R, Akiyama R, Tissot N, Milosavljevic S, Yamazaki M, Wyler M, Arongaus AB, Podolec R, **Sato Y**, Widmer A, Ulm R*, Shimizu KK*. (2024) UV RESISTANCE LOCUS 8-mediated UV-B response is required alongside CRYPTOCHROME 1 for plant survival in sunlight under field conditions. ***Plant and Cell Physiology*** 65:35-48. https://doi.org/10.1093/pcp/pcad113
6. The Herbivory Variability Network. (2023) Plant size, latitude, and phylogeny explain variability in herbivory. ***Science*** 382:679-683. https://doi.org/10.1126/science.adh8830 [a consortium paper including Y. Sato as one of 191 co-authors]
7. Xu C†#, **Sato Y**†*, Yamazaki M, Brasser M, Barbour MA, Bascompte J, Shimizu KK*. (2023) Genome-wide association study of aphid abundance highlights a locus affecting plant growth and flowering in *Arabidopsis thaliana*. ***Royal Society Open Science*** 10:230399. https://doi.org/10.1098/rsos.230399
8. **Sato Y***, Takahashi Y, Xu C#, Shimizu KK*. (2023) Detecting frequency-dependent selection through the effects of genotype similarity on fitness components. ***Evolution*** 77:1145–1157. https://doi.org/10.1093/evolut/qpad028
9. Shimizu-Inatsugi R*, Morishima A, Mourato B, Shimizu KK, **Sato Y**. (2023) Phenotypic variation of a new synthetic allotetraploid *Arabidopsis kamchatica* enhanced in natural environment. ***Frontiers in Plant Science*** 13:1058522 https://doi.org/10.3389/fpls.2022.1058522
10. Takimoto H†*, **Sato Y**†, Nagano AJ, Shimizu KK, Kanagawa A. (2021) Using a two-stage convolutional neural network to rapidly identify tiny herbivorous beetles in the field. ***Ecological Informatics*** 66:101466 https://doi.org/10.1016/j.ecoinf.2021.101466
11. **Sato Y**, Yamamoto E, Shimizu KK*, Nagano AJ*. (2021) Neighbor GWAS: incorporating neighbor genotypic identity into genome-wide association studies of field herbivory. ***Heredity*** 126(4):597–614. https://doi.org/10.1038/s41437-020-00401-w
12. **Sato Y**†*, Takeda K†, Nagano AJ*. (2021) Neighbor QTL: an interval mapping method for quantitative trait loci underlying plant neighborhood effects. ***G3; Genes|Genomes|Genetics*** 11(2):jkab017. https://doi.org/10.1093/g3journal/jkab017
13. Shimizu-Inatsugi R*, Milosavljevic S, Shimizu KK, Schaepman-Strub G, Tanoi K, **Sato Y.** (2021) Metal accumulation and its effect on leaf herbivory in an allopolyploid species *Arabidopsis kamchatica* inherited from a diploid hyperaccumulator *A. halleri*. ***Plant Species Biology*** 36(2):208–217. https://doi.org/10.1111/1442-1984.12304
14. **Sato Y**, Tezuka A, Kashima M, Deguchi A, Shimizu-Inatsugi R, Yamazaki M, Shimizu KK*, Nagano AJ*. (2019) Transcriptional variation in glucosinolate biosynthetic genes and inducible responses to aphid herbivory on field-grown *Arabidopsis thaliana*. ***Frontiers in Genetics*** 10:787. https://doi.org/10.3389/fgene.2019.00787
15. **Sato Y**, Shimizu-Inatsugi R, Yamazaki M, Shimizu KK*, Nagano AJ*. (2019) Plant trichomes and a single gene *GLABRA1* contribute to insect community composition on field-grown *Arabidopsis thaliana*. ***BMC Plant Biology*** 19:163. https://doi.org/10.1186/s12870-019-1705-2
16. Nakadai R†*, Hashimoto K†, Iwasaki T, **Sato Y.** (2018) Geographical co-occurrence of butterfly species: the importance of niche filtering by host plant species. ***Oecologia*** 186(4):995–1005. https://doi.org/10.1007/s00442-018-4062-1
17. **Sato Y**†*, Ito K†, Kudoh H. (2017) Optimal foraging by herbivores maintains polymorphism in defence in a natural plant population. ***Functional Ecology*** 31(12):2233-2243. https://doi.org/10.1111/1365-2435.12937
18. **Sato Y***, Kudoh H. (2017) Herbivore-mediated interaction promotes the maintenance of trichome dimorphism through negative frequency-dependent selection. ***The American Naturalist*** 190(3):E67-E77. https://doi.org/10.1086/692603

19. **Sato Y***, Kudoh H. (2017) Fine-scale frequency differentiation along a herbivory gradient in the trichome dimorphism of a wild *Arabidopsis*. ***Ecology and Evolution*** 7(7):2133-2141. https://doi.org/10.1002/ece3.2830
20. **Sato Y***, Kudoh H. (2016) Presence of substitute diets alters plant resistance to specialist and generalist herbivores: a meta-analysis. ***Ecosphere*** 7(11):e01446 https://doi.org/10.1002/ecs2.1446
21. **Sato Y***, Kudoh H. (2016) Associational effects against a leaf beetle mediate a minority advantage in defense and growth between hairy and glabrous plants. ***Evolutionary Ecology*** 30(1):137-154. https://doi.org/10.1007/s10682-015-9809-0
22. **Sato Y***, Sato S. (2015) Spring temperature predicts the long-term molting phenology of two cicadas, *Cryptotympana facialis* and *Graptopsaltria nigrofuscata* (Hemiptera: Cicadidae). ***Annals of the Entomological Society of America*** 108(4):494-500. https://doi.org/10.1093/aesa/sav036
23. **Sato Y***, Kudoh H. (2015) Tests of associational defence provided by hairy plants for glabrous plants of *Arabidopsis halleri* subsp. *gemmifera* against insect herbivores. ***Ecological Entomology*** 40(3):269-279. https://doi.org/10.1111/een.12179
24. **Sato Y**, Kudoh H*. (2014) Fine-scale genetic differentiation of a temperate herb: relevance of local environments and demographic change. ***AoB PLANTS*** 6:plu070. https://doi.org/10.1093/aobpla/plu070
25. **Sato Y***, Kawagoe T, Sawada Y, Hirai MY, Kudoh H. (2014) Frequency-dependent herbivory by a leaf beetle, *Phaedon brassicae*, on hairy and glabrous plants of *Arabidopsis halleri* subsp. *gemmifera*. ***Evolutionary Ecology*** 28(3):545-559. https://doi.org/10.1007/s10682-013-9686-3
26. **Sato Y***, Kudoh H. (2013) Relative strength of phenotypic selection on the height and number of flowering-stalks in the rosette annual *Cardamine hirsuta* (Brassicaceae). ***Journal of Ecology and Environment*** 36(3):151-158. https://doi.org/10.5141/ecoenv.2013.151
27. **Sato Y***, Takakura KI, Nishida S, Nishida T. (2013) Dominant occurrence of cleistogamous flowers of *Lamium amplexicaule* in relation to the nearby presence of an alien congener *L. purpureum*. ***ISRN Ecology*** Article ID:476862. http://dx.doi.org/10.1155/2013/476862

## Peer-reviewed monographs

1. **Sato Y***†, Wuest SE*. (2025) The genetics of plant–plant interactions and their cascading effects on agroecosystems – from model plants to applications. 66(4):477-489. ***Plant and Cell Physiology*** https://doi.org/10.1093/pcp/pcae092
2. **Sato Y*.** (2018) Associational effects and the maintenance of polymorphism in plant defense against herbivores: review and evidence. ***Plant Species Biology*** 33(2):91-108. https://doi.org/10.1111/1442-1984.12201
3. Sakata Y*, **Sato Y.** (2017) Evolutionary ecology of plant defense: integrating different spatial scales within and among species (in Japanese). ***Japanese Journal of Ecology*** 67(3):287-306. https://doi.org/10.18960/seitai.67.3_287

## Peer-reviewed conference proceedings

1. Kuzuhara H, Takimoto H, **Sato Y**, Kanagawa A. (2020) Insect pest detection and identification method based on deep learning for realizing a pest control system. ***Proceedings of the Society of Instrument and Control Engineers (SICE) 2020 Annual Conference 2020***. pp. 709–714. https://doi.org/10.23919/SICE48898.2020.9240458

## Contributions to books

1. **Sato Y**, Akram I#. (2026). Chapter six - Extending genomic prediction from individuals to a population and community. In: ***Advances in Ecological Research*** vol. 74, pp. 153–189. https://doi.org/10.1016/bs.aecr.2026.02.001
2. **Sato Y**. (2024) Chapter 7: The maintenance of trichome dimorphism through associational effects: a case study of *Arabidopsis halleri* and a leaf beetle (in Japanese). In: Ecology of Plant Defense Against Herbivores (Y. Sakata and T. Tsunoda eds), ***Species Biology Series***, Bun-ichi Sogo Shuppan, Tokyo,

Japan. ISBN:978-4-8299-6211-4.
3. **Sato Y**, Muranaka T. (eds). (2023) Trans-Scale Biology of Plants: Integrating genomics, phenomics, and ecology to unveil biodiversity (in Japanese). ***Species Biology Series***, Bun-ichi Sogo Shuppan, Tokyo, Japan. ISBN:978-4-8299-6210-7.

## Patents and licenses

*None*

## Computer programs

1. **Sato Y***, Yamamoto E, Shimizu KK, Nagano AJ. (2021) 'rNeighborGWAS'. R package available at the Comprehensive R Archive Network (CRAN), https://cran.r-project.org/package=rNeighborGWAS
2. **Sato Y***, Takeda K, Nagano AJ. (2021) 'rNeighborQTL'. R package available at CRAN, https://cran.r-project.org/package=rNeighborQTL

*And the other source codes for published research, personal utilities, and teaching materials are available at the GitHub repository, https://github.com/yassato*

## Conference contributions

1. **Sato Y**. Linking frequency-dependent selection and overyielding through genome-wide association study of plant neighborhood effects. ***Evolution 2025 Virtual Conference*** (29 May 2025, online talk)
2. **Sato Y**, Shimizu-Inatsugi R, Takeda K, Nagano AJ, Shimizu KK. Genomic prediction of neighbor effects identifies key genotype pairs responsible for reduced herbivory in mixed planting. ***72nd Annual Meeting of Ecological Society of Japan***, Sapporo, Hokkaido, Japan (17 March 2025, talk)
3. **Sato Y**, Takahashi Y, Xu C, Shimizu KK. Detecting frequency-dependent selection through the effects of genotype similarity on fitness components. ***biology$^{24}$: The largest conference of organismal biology in Switzerland***, ETH Zurich, Switzerland (19 January 2024, poster)
4. **Sato Y**, Shimizu-Inatsugi R, Takeda K, Nagano AJ, Shimizu KK. Genome-wide neighbor effects predict genotype pairs reducing herbivory in mixed planting. ***Biodiversity Convention - From Science to Implementation***, Monte Verita, Ascona, Switzerland (27 June 2023, talk)
5. **Sato Y**, Shimizu-Inatsugi R, Takeda K, Nagano AJ, Shimizu KK. Keystone pairs of *Arabidopsis* accessions increase plant resistance to field herbivory. In Concurrent 15: *Arabidopsis* relatives from laboratories to natural fields (organized by Prof. K.K. Shimizu and H. Kudoh). ***The 33rd International Conference on Arabidopsis Research (ICAR2023)***, Chiba, Japan (7 June 2023, talk)
6. **Sato Y**, Shimizu-Inatsugi R, Takeda K, Nagano AJ, Shimizu KK. Keystone genotype pairs increase plant resistance to herbivory. ***Gordon Research Conference: Plant-Herbivore Interaction 2023***, Ventura, CA, USA (2 March 2023, poster)
7. **Sato Y**, Shimizu-Inatsugi R, Takeda K, Nagano AJ, Shimizu KK. Keystone genotype pairs increase population-level resistance to herbivory. ***SwissPLANT 2023 Symposium***, Les Diablerets, Switzerland (23 January 2023, talk)
8. **Sato Y**, Shimizu-Inatsugi R, Nagano AJ, Shimizu KK. Neighbor GWAS: incorporating neighbor genotypic identity in genome-wide association study of field-grown *Arabidopsis thaliana*. 'From place to space - Tracing the spatial dimension of plant sciences' by ***Plant Science Center Symposium 2022***, P30, ETH Zurich, Switzerland (7 December 2022, poster)
9. **Sato Y**, Takahashi Y, Xu C, Shimizu KK. Detecting frequency-dependent selection using a genetic marker regression of fitness components. In: S24 'Progress and Prospects in Adaptation Genomics'. ***The 2022 Congress of the European Society for Evolutionary Biology***, Prague, Czech Republic (18 August 2022, reviewed talk)
10. **Sato Y**, Nagano AJ, Shimizu KK. Genomics of within-species mixed planting against insect herbivores. In: MON3-b 'From the species to the individual: investigating plant diversity on the scale that matters most' (organized by Drs. P. Villa and M.C. Schuman). ***World Biodiversity Forum 2022***, Davos, Switzerland. (29 June 2022, reviewed talk) https://www.worldbiodiversityforum.org/

11. Kuzuhara H, Takimoto H, **Sato Y**, Kanagawa A. Insect pest detection and identification method based on deep learning for realizing a pest control system. ***The Society of Instrument and Control Engineers (SICE) Annual Conference 2020***. Chiang Mai, Thailand [changed into an online meeting due to COVID-19] (25 September 2020, talk)
12. **Sato Y**. Using personal legacy data to reveal the molting phenology of Japanese cicadas. In: S19 'Ongoing Cicada Research in East Asia' (organized by Prof. J. Yikweon), ***The 8th EAFES International Congress***, Nagoya, Japan (April 2018, invited talk)

*And the other 7 English posters and 37 contributions in Japanese (5 invited talks, 10 talks and 23 posters)*

## Outreach activities

1. "Exhibition of *Arabidopsis thaliana* worldwide" at the Faculty of Environmental Earth Science, Hokkaido University, Japan (8 June 2025, in Japanese)
2. Teaching assistant for making a botanical specimen at Japanese School in Zurich (Japanische Schule in Zuerich), Uster, Switzerland. (1 July 2023, in Japanese)
3. **Sato Y**, Sato S. (2016) 12-years cicada monitoring. ***Proceedings of the Natural History Museum Mt. Fujiwara*** 38:1–5 (in Japanese)
   *-> Plain-text Japanese summary of Sato and Sato (2015) Annals of the Entomological Society of America 108(4):494–500.*

## Preprint

1. Pérez-López CB, White F, Abonnenc N, L'Anglais-Landry S, Milot É, MacDonald AM, Jacques PÉ, **Sato Y**, Barbour MA*. (2026). Epistasis and an oligogenic architecture underlie the strength of a plant-herbivore interaction. ***bioRxiv*** https://doi.org/10.64898/2026.09.14.751214
2. Tomita A, Maeda T, Mori-Moriyama N, Nomura Y, Kurita Y, Kashima M, Betsuyaku S, **Sato Y***, Nagano AJ*. (2026). Transcriptomic biomarkers reveal jasmonic and salicylic acid state under field herbivory. ***bioRxiv*** https://doi.org/10.1101/2025.05.29.656841
3. **Sato Y***, Hamazaki K*. (2026) Dissecting oligogenic and polygenic indirect genetic effects through the lens of neighbor genotypic identity. ***eLife*** https://doi.org/10.7554/eLife.111650 [version 1]

## Others

1. Tsuchimatsu T*, Sasaki E, **Sato Y**, Castric V. (2025) Ecology and evolution: genetic and molecular dynamics of plants in nature. ***Plant and Cell Physiology*** 66(4):427–430. https://doi.org/10.1093/pcp/pcaf035 [Editorial preface for the special issue 'Ecology and Evolution']